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CODETANTRA LAB PRACTICAL 1

The screenshot shows the CODETANTRA lab interface for the problem "1.1.1. Calculate Momentum". The problem description states: "Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum p is calculated using the formula: $p = m \times v$ where: m is the mass of the object (in kilograms), v is the velocity of the object (in meters per second)."

Input Format:

- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

Output Format:

- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

The code editor shows the following Python code:

```
1 mass=float(input())
2 velocity=float(input())
3 mom=mass*velocity
4 print(f"{mom:.2f}kgm/s")
```

The test results show that 2 out of 2 shown test case(s) passed and 2 out of 2 hidden test case(s) passed. The average time is 0.019 s and the maximum time is 0.029 s.

Test Case	Expected output	Actual output
Test case 1	5.0 10.0 50.00kg/s	5.0 10.0 50.00kg/s
Test case 2		

The screenshot shows the CODETANTRA lab interface for the problem "1.1.2. Conditional Calculation Based on the Number of Digits". The problem description states: "Write a Python program that accepts an integer n as input. Depending on the number of digits in n . Constraints: $1 \leq n \leq 999$ ".

Input Format:

- The input consists of a single integer n .

Output Format:

- If n is a single-digit number, print its square.
- If n is a two-digit number, print its square root (rounded to two decimal places).
- If n is a three-digit number, print its cube root (rounded to two decimal places).
- Else print "Invalid".

The code editor shows the following Python code:

```
1 import math
2 n=int(input())
3 if n>=1 and n<=9:
4     print(n*n)
5 elif n>=10 and n<=99:
6     print(f"{math.sqrt(n):.2f}")
7 elif n>=100 and n<=999:
8     print(f"{math.cbrt(n):.2f}")
9 else:
10    print("Invalid")
```

The test results show that 4 out of 4 shown test case(s) passed and 3 out of 3 hidden test case(s) passed. The average time is 0.004 s and the maximum time is 0.006 s.

Test Case	Expected output	Actual output
Test case 1	9 81	9 81
Test case 2		
Test case 3		

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1.1.3. Days between Two Dates

Write a Python program to calculate number of days between two dates.
Input Format:

- The first line contains the first date in the format `YYYY-MM-DD`.
- The second line contains the second date in the format `YYYY-MM-DD`.

Output Format:

- An integer representing the number of days between the given dates.

Note:

- The first date should always be considered earlier than the second date.

Sample Test Cases

noOfDays...

```
1 from datetime import datetime
2
3 date1_str = input().strip()
4 date2_str = input().strip()
5
6 date1 = datetime.strptime(date1_str, "%Y-%m-%d")
7 date2 = datetime.strptime(date2_str, "%Y-%m-%d")
8
9 difference = date2 - date1
10 print(difference.days)
```

Average time0.007 s7.50 msMaximum time0.016 s16.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 116 ms

Expected output
2016-07-02
2014-11-02
123

Actual output
2014-07-02
2014-11-02
123

Test case 26 ms

TerminalTest cases

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1.1.4. Reverse a Number

Write a Python program to reverse the digits of the number and print the reversed number.
Input Format:

- The input is a single integer.

Output Format:

- The output prints a single integer, which is the reversed number.

Sample Test Cases

reverseN...

```
1 n = (int(input()))
2 rev = int(str(n)[::-1])
3 print (rev)
```

Average time0.004 s3.60 msMaximum time0.005 s5.00 ms

2 out of 2 shown test case(s) passed
3 out of 3 hidden test case(s) passed

Test case 15 ms

Expected output
5367
7635

Actual output
5367
7635

Test case 23 ms

TerminalTest cases

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1.1.5. Multiplication Table

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.
Input Format:

- The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

- Print the multiplication table for the given number in the format:
<number> x <multiplier> = <result>

Note:

- The character "x" is used in the output to represent multiplication.
- Refer to the visible test-cases for more clarity.

Sample Test Cases

multipl...

```
1 n=int(input())
2 for i in range(1,11):
3     print(f"{n} x {i} = {n*i}")
4
5
6
```

Average time0.003 s2.50 msMaximum time0.004 s4.00 ms

2 out of 2 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 1

Expected output

Actual output

8

8 x 1 = 8

8 x 2 = 16

8 x 3 = 24

8 x 4 = 32

8 x 5 = 40

TerminalTest cases

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1.2.1. Pass or Fail

Write a Python program that accepts the number of courses and the marks of a student in those courses.
The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer n , the number of courses.
- The second input will be n integers representing the marks of the student in each of the n courses, separated by a space.

Output Format:

- If the student passes all courses:
 - Print the aggregate percentage (formatted to two decimal places).
 - Print the grade based on the aggregate percentage.
- If the student fails any course (marks < 40 in any course), print:
 - "Fail".

Note:

- Aggregate Percentage: Average performance across all courses, calculated by summing all marks and dividing by the number of courses.

Sample Test Cases

passorFa...

```
1 num_courses=int(input())
2 marks=list(map(int,input().split()))
3 has_failed=any(mark<40 for mark in marks)
4 if has_failed:
5     print("Fail")
6 else:
7     aggregate=sum(marks)/num_courses
8     if aggregate>75:
9         grade="Grade: Distinction"
10    elif aggregate>60:
11        grade="Grade: First Division"
12    elif aggregate>50:
13        grade="Grade: Second Division"
14    elif aggregate>40:
15        grade="Grade: Third Division"
```

Average time0.005 s4.90 msMaximum time0.007 s7.00 ms

5 out of 5 shown test case(s) passed5 out of 5 hidden test case(s) passed

Test case 1

Expected output

Actual output

4

55 65 78 89

Aggregate Percentage: 71.75

Grade: First Division

Test case 2

TerminalTest cases

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1.2.2. Fibonacci Series

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Write a Python program that uses recursion to print the first n terms of the Fibonacci series.

Input Format:

- A single integer n representing the number of terms to generate.

Output Format:

- A single line containing the first n terms of the Fibonacci sequence, separated by spaces.

Sample Test Cases

fibonacci...

```
1 #def fibonacci(n):
2 def fibonacci(n):
3     if n<=0:
4         return 0
5
6     elif n==1:
7         return 0
8     elif n==2:
9         return 1
10    else:
11        return fibonacci( n-1 ) + fibonacci(n-2)
12
13
14 n = int(input())
15 for i in range(1, n + 1):
```

Average time0.017 s16.50 ms

Maximum time0.024 s24.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 117 ms

Expected output

0 1 1 2 3

Actual output

0 1 1 2 3

Test case 211 ms

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1.2.3. Pattern - 1

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Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format

- The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.

Sample Test Cases

rightangl...

```
1 n=int(input())
2 for i in range(1,n+1):
3     print('*'*i)
```

Average time0.003 s3.33 ms

Maximum time0.006 s6.00 ms

2 out of 2 shown test case(s) passed

4 out of 4 hidden test case(s) passed

Test case 16 ms

Expected output

*
* *
* * *
* * * *
* * * * *

Actual output

*
* *
* * *
* * * *
* * * * *

TerminalTest cases

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1.2.4. Pattern - 2

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Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

The input is an integer, representing the number of rows in the pattern.

Output Format:

The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Note:

Refer to the displayed test cases for the sample pattern.

Sample Test Cases

numberP...

```
1 # Type Content here...
2 n=int(input())
3 for i in range(1,n+1):
4     for j in range(1, i+1):
5         print(j, end=" ")
6     print()
7
8
```

Average time

0.004 s

4.90 ms

Maximum time

0.007 s

7.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

Expected output

Actual output

1

1

1 2

1 2

1 2 3

1 2 3

1 2 3 4

1 2 3 4

1 2 3 4 5

1 2 3 4 5

Terminal

Test cases

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Reset

Submit

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